

53. A simple pendulum is being used to determine the value of gravitational acceleration g at a certain place. The length of the pendulum is 25.0 cm and a stopwatch with 1 s resolution measures the time taken for 40 oscillation to be 50 s. The accuracy in g is

[JEE (Main)-2020]

- (1) 3.40% (2) 2.40%
(3) 5.40% (4) 4.40%

54. A physical quantity z depends on four observables a , b , c and d , as $z = \frac{a^2 b^2}{\sqrt{c} d^3}$. The

percentages of error in the measurement of a , b , c and d are 2%, 1.5%, 4% and 2.5% respectively. The percentage of error in z is

[JEE (Main)-2020]

- (1) 13.5% (2) 14.5%
(3) 16.5% (4) 12.25%

55. A wire of 1Ω has a length of 1 m. It is stretched till its length increases by 25%. The percentage change in resistance to the nearest integer is

[JEE (Main)-2021]

- (1) 76% (2) 56%
(3) 12.5% (4) 25%

56. In the experiment of Ohm's law, a potential difference of 5.0 V is applied across the end of a conductor of length 10.0 cm and diameter of 5.00 mm. The measured current in the conductor is 2.00 A. The maximum permissible percentage error in the resistivity of the conductor is:

[JEE (Main)-2021]

- (1) 3.0 (2) 3.9
(3) 7.5 (4) 8.4

57. Two resistors $R_1 = (4 \pm 0.8) \Omega$ and $R_2 = (4 \pm 0.4) \Omega$ are connected in parallel. The equivalent resistance of their parallel combination will be:

[JEE (Main)-2021]

- (1) $(4 \pm 0.4) \Omega$ (2) $(2 \pm 0.4) \Omega$
(3) $(2 \pm 0.3) \Omega$ (4) $(4 \pm 0.3) \Omega$

58. If $Z = \frac{A^2 B^3}{C^4}$, then the relative error in Z will be

[JEE (Main)-2022]

- (1) $\frac{\Delta A}{A} + \frac{\Delta B}{B} + \frac{\Delta C}{C}$
(2) $\frac{2\Delta A}{A} + \frac{3\Delta B}{B} - \frac{4\Delta C}{C}$

$$(3) \frac{2\Delta A}{A} + \frac{3\Delta B}{B} + \frac{4\Delta C}{C}$$

$$(4) \frac{\Delta A}{A} + \frac{\Delta B}{B} - \frac{\Delta C}{C}$$

59. A silver wire has a mass (0.6 ± 0.006) g, radius (0.5 ± 0.005) mm and length (4 ± 0.04) cm. The maximum percentage error in the measurement of its density will be

[JEE (Main)-2022]

- (1) 4% (2) 3%
(3) 6% (4) 7%

60. A torque meter is calibrated to reference standards of mass, length and time each with 5% accuracy. After calibration, the measured torque with this torque meter will have net accuracy of

[JEE (Main)-2022]

- (1) 15% (2) 25%
(3) 75% (4) 5%

61. If the radius of earth shrinks by 2% while its mass remains same. The acceleration due to gravity on the earth's surface will approximately

[JEE (Main)-2022]

- (1) Decrease by 2% (2) Decrease by 4%
(3) Increase by 2% (4) Increase by 4%

62. An aluminium wire is stretched to make its length, 0.4% larger. The percentage change in resistance is:

[JEE (Main)-2022]

- (1) 0.4% (2) 0.2%
(3) 0.8% (4) 0.6%

63. The maximum error in the measurement of resistance, current and time for which current flows in an electrical circuit are 1%, 2% and 3% respectively. The maximum percentage error in the detection of the dissipated heat will be

[JEE (Main)-2022]

- (1) 2 (2) 4
(3) 6 (4) 8

64. A cylindrical wire, of mass (0.4 ± 0.01) g has length (8 ± 0.04) cm and radius (6 ± 0.03) mm. The maximum error in its density will be

[JEE (Main)-2023]

- (1) 3.5% (2) 5%
(3) 1% (4) 4%

65. The energy of a system is given as $E(t) = \alpha^3 e^{-\beta t}$, where t is the time and $\beta = 0.3 \text{ s}^{-1}$. The errors in the measurement of α and t are 1.2% and 1.6%, respectively. At $t = 5$ s, maximum percentage error in the energy is:

[JEE (Main)-2025]

- (1) 6% (2) 4%
(3) 8.4% (4) 11.6%

[Miscellaneous]

66. A huge circular arc of length 4.4 ly subtends an angle '4s' at the centre of the circle. How long it would take for a body to complete 4 revolution if its speed is 8 AU per second?

Given: 1 ly = 9.46×10^{15} m
1 AU = 1.5×10^{11} m

[JEE (Main)-2021]

- (1) 3.5×10^6 s (2) 4.5×10^{10} s
(3) 4.1×10^8 s (4) 7.2×10^8 s

67. Consider two physical quantities A and B related to each other as $E = \frac{B-x^2}{At}$ where E, x and t have dimensions of energy, length and time respectively. The dimension of AB is

[JEE (Main)-2024]

- (1) $L^{-2}M^1T^0$ (2) $L^{-2}M^{-1}T^1$
(3) $L^0M^{-1}T^1$ (4) $L^2M^{-1}T^1$

68. If the percentage errors in measuring the length and the diameter of a wire are 0.1% each. The percentage error in measuring its resistance will be

[JEE (Main)-2024]

- (1) 0.2% (2) 0.144%
(3) 0.3% (4) 0.1%

Numerical Value Based Questions

69. The density of a solid metal sphere is determined by measuring its mass and its diameter. The maximum error in the density of the sphere is $\left(\frac{x}{100}\right)\%$. If the relative errors in measuring the mass and the diameter are 6.0% and 1.5% respectively, the value of x is _____.

[JEE (Main)-2020]

- 70*. The resistance $R = \frac{V}{I}$, where $V = (50 \pm 2)$ V and $I = (20 \pm 0.2)$ A. The percentage error in R is 'x' %.

The value of 'x' to the nearest integer is _____.

[JEE (Main)-2021]

71. The radius of a sphere is measured to be (7.50 ± 0.85) cm. Suppose the percentage error in its volume is x.

The value of x, to the nearest x, is _____.

[JEE (Main)-2021]

72. The acceleration due to gravity is found upto an accuracy of 4% on a planet. The energy supplied to a simple pendulum of known mass 'm' to undertake oscillations of time period T is being estimated. If time period is measured to an accuracy of 3%, the accuracy to which E is known as _____ %.

[JEE (Main)-2021]

73. For $z = a^2x^3y^{\frac{1}{2}}$, where 'a' is a constant. If percentage error in measurement of 'x' and 'y' are 4% and 12%, respectively, then the percentage error for 'z' will be _____ %.

[JEE (Main)-2022]

74. A student in the laboratory measures thickness of a wire using screw gauge. The readings are 1.22 mm, 1.23 mm, 1.19 mm and 1.20 mm. The percentage error is $\frac{x}{121}\%$. The value of x is _____.

[JEE (Main)-2022]

75. In an experiment to find acceleration due to gravity (g) using simple pendulum, time period of 0.5 s is measured from time of 100 oscillations with a watch of 1 s resolution. If measured value of length is 10 cm known to 1 mm accuracy, the accuracy in the determination of g is found to be x%. The value of x is _____.

[JEE (Main)-2022]

76. The length of a given cylindrical wire is increased to double of its original length. The percentage increase in the resistance of the wire will be _____ %.

[JEE (Main)-2022]

77. A physical quantity Q is related to four observables a, b, c, d as follows : $Q = \frac{ab^4}{cd}$ where, $a = (60 \pm 3)$ Pa; $b = (20 \pm 0.1)$ m; $c = (40 \pm 0.2)$ Nsm⁻² and $d = (50 \pm 0.1)$ m, then the percentage error in Q is $\frac{x}{1000}$

[JEE (Main)-2025]

78. In a measurement, it is asked to find modulus of elasticity per unit torque applied on the system. The measured quantity has dimension of $[M^aL^bT^c]$. If $b = 3$, then value of c is _____.

[JEE (Main)-2025]

SECTION - G (PREVIOUS YEARS QUESTIONS)

Physical World, Units and Measurement 35

Only One Option Correct Type Questions

1. A student performs an experiment to determine the Young's modulus of a wire, exactly 2 m long, by Searle's method. In a particular reading, the student measures the extension in the length of the wire to be 0.8 mm with an uncertainty of ± 0.05 mm at load of exactly 1.0 kg. The student also measures the diameter of the wire to be 0.4 mm with an uncertainty of ± 0.01 mm. Take $g = 9.8 \text{ m/s}^2$ (exact). The Young's modulus obtained from the reading is
 (A) $(2.0 \pm 0.3) \times 10^{11} \text{ N/m}^2$
 (B) $(2.0 \pm 0.2) \times 10^{11} \text{ N/m}^2$
 (C) $(2.0 \pm 0.1) \times 10^{11} \text{ N/m}^2$
 (D) $(2.0 \pm 0.05) \times 10^{11} \text{ N/m}^2$
 [IIT-JEE-2007 (Paper-2)]
2. A person measures the depth of a well by measuring the time interval between dropping a stone and receiving the sound of impact with the bottom of the well. The error in his measurement of time is $\delta T = 0.01$ seconds and he measures the depth of the well to be $L = 20$ meters. Take the acceleration due to gravity $g = 10 \text{ ms}^{-2}$ and the velocity of sound is 300 ms^{-1} . Then the fractional error in the measurement, $\delta L/L$, is closest to
 (A) 1%
 (B) 5%
 (C) 3%
 (D) 0.2%
 [JEE (Adv)-2017 (Paper-2)]
3. Young's modulus of elasticity Y is expressed in terms of three derived quantities, namely, the gravitational constant G , Planck's constant h and the speed of light c , as $Y = c^\alpha h^\beta G^\gamma$. Which of the following is the correct option?
 (A) $\alpha = 7, \beta = -1, \gamma = -2$
 (B) $\alpha = -7, \beta = -1, \gamma = -2$
 (C) $\alpha = 7, \beta = -1, \gamma = 2$
 (D) $\alpha = -7, \beta = 1, \gamma = -2$
 [JEE (Adv)-2023 (Paper-2)]
4. A dimensionless quantity is constructed in terms of electronic charge e , permittivity of free space ϵ_0 , Planck's constant h , and speed of light c . If the dimensionless quantity is written as $e^\alpha \epsilon_0^\beta h^\gamma c^\delta$ and n is a non-zero integer, then $(\alpha, \beta, \gamma, \delta)$ is given by
 (A) $(2n, -n, -n, -n)$
 (B) $(n, -n, -2n, -n)$
 (C) $(n, -n, -n, -2n)$
 (D) $(2n, -n, -2n, -2n)$
 [JEE (Adv)-2024 (Paper-1)]
5. A temperature difference can generate e.m.f. in some materials. Let S be the e.m.f. produced per unit temperature difference between the ends of a wire, σ the electrical conductivity and κ the thermal conductivity of the material of the wire. Taking M, L, T, I and K as dimensions of mass, length, time, current and temperature, respectively, the dimensional formula of the quantity $Z = \frac{S^2 \sigma}{\kappa}$ is
 (A) $[M^0 L^0 T^0 P^0 K^0]$
 (B) $[M^0 L^0 T^0 P^0 K^{-1}]$
 (C) $[M^1 L^2 T^{-2} I^{-1} K^{-1}]$
 (D) $[M^1 L^2 T^{-4} I^{-1} K^{-1}]$
 [JEE (Adv)-2025 (Paper-2)]

One or More Option(s) Correct Type Questions

6. Sometimes it is convenient to construct a system of units so that all quantities can be expressed in terms of only one physical quantity. In one such system, dimensions of different quantities are given in terms of a quantity X as follows: [position] = $[X^a]$; [speed] = $[X^b]$; [acceleration] = $[X^c]$; [linear momentum] = $[X^d]$; [force] = $[X^e]$. Then
 (A) $\alpha + p = 2\beta$
 (B) $p + q - r = \beta$
 (C) $p - q + r = \alpha$
 (D) $p + q + r = \beta$
 [JEE (Adv)-2020 (Paper-1)]

36 Physical World, Units and Measurement

7. Length, breadth and thickness of a strip having a uniform cross section are measured to be 10.5 cm, 0.05 mm, and 6.0 μm , respectively. Which of the following option(s) give(s) the volume of the strip in cm^3 with correct significant figures :

(A) 3.2×10^{-5}
(C) 3.0×10^{-5}

(B) 32.0×10^{-6}
(D) 3×10^{-5}

[JEE (Adv)-2025 (Paper-1)]



Matrix-Match / Matrix Based Type Questions

8. Match List-I with List-II and select the correct answer using the codes given below the lists:

[JEE (Adv)-2013 (Paper-2)]



	List-I		List-II
(P)	Boltzmann constant	(1)	$[\text{ML}^2\text{T}^{-1}]$
(Q)	Coefficient of viscosity	(2)	$[\text{ML}^{-1}\text{T}^{-1}]$
(R)	Planck constant	(3)	$[\text{MLT}^{-3}\text{K}^{-1}]$
(S)	Thermal conductivity	(4)	$[\text{ML}^2\text{T}^{-2}\text{K}^{-1}]$

Codes :

	P	Q	R	S
(A)	3	1	2	4
(B)	3	2	1	4
(C)	4	2	1	3
(D)	4	1	2	3

Integer / Numerical Value Type Questions

9. The energy of a system as a function of time t is given as $E(t) = A^2 \exp(-\alpha t)$, where $\alpha = 0.2 \text{ s}^{-1}$. The measurement of A has an error of 1.25%. If the error in the measurement of time is 1.50%, the percentage error in the value of $E(t)$ at $t = 5 \text{ s}$ is

[JEE (Adv)-2015 (Paper-2)]



10. In a particular system of units, a physical quantity can be expressed in terms of the electric charge e , electron mass m_e , Planck's constant h , and Coulomb's constant

$k = \frac{1}{4\pi\epsilon_0}$, where ϵ_0 is the permittivity of vacuum. In terms of these physical constants, the dimension of the magnetic field is $[B] = [e]^\alpha [m_e]^\beta [h]^\gamma [k]^\delta$. The value of $\alpha + \beta + \gamma + \delta$ is _____.

[JEE (Adv)-2022 (Paper-2)]



11. The dimensions of a cone are measured using a scale with a least count of 2 mm. The diameter of the base and the height are both measured to be 20.0 cm. The maximum percentage error in the determination of the volume is _____.

[JEE (Adv)-2024 (Paper-2)]

